COURSE TITLE	**DIGITAL DESIGN USING VERILOG**
COURSE CODE	**01CT1619**
COURSE CREDITS	**4**

Objective:

1 This course will introduce the students to the Verilog Hardware Description Language (HDL). Students will learn to model the hardware using hardware description language.

Course Outcomes: After completion of this course, student will be able to:

1 Understand the Verilog Hardware Description Language
2 Model the combinational and sequential circuits using Verilog
3 Verify the digital design with simulation
4 Analyze the different Verilog modeling style
5 Interpret the data sheets of CPLD and FPGA

Pre-requisite of course: Digital Electronics

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	2	50	30	20	25	25

Contents : Unit	Topics	Contact Hours
1	**Digital Circuit Design Flow** Introduction to digital circuit design flow, VLSI design flow, Behavioral design, data path design, logic design, physical design, manufacturing. VLSI design style, FPGA based design, standard cell based design, Full custom design	4
2	**Introduction to Verilog** Concept of module , Data types in Verilog, Features of Verilog, Verilog operators	3
3	**Verilog Modeling** Behavioral modeling and structural modeling, Explanation of modeling with different examples, modeling 16x1 multiplexer, modeling 16 bit adder, Switch level modeling	6
4	**Verilog Description Styles** Data flow and behavioral(procedure statements), Example of decoder and D-latch, Various examples using blocking and non blocking assignment, Analysing that how the modelling style influences the simulator and synthesis tool.	7

MR. CHANDRASINH DHIRUBHA PARMAR
Digitally signed by (Name of HOD)

DR. RAJENDRASINH BAHADURSINH JADEJA
Digitally signed by (Name of Dean/ Principal)

Contents : Unit	Topics	Contact Hours
5	**Verilog Test benches** Writing Verilog test benches, write test bench for combinational design, write test bench for sequential design, generating test vectors	4
6	**Modeling Finite State Machine** Moore Machine, Mealy Machine, Examples	6
7	**Data path and Controller design** Data path design includes functional circuits, registers, multiplexers, bus, adders, multipliers, counters etc., Controller design includes Finite State Machine and provide control to the data path,, Example to explain above concept	3
8	**Modeling Memory and register banks** Concepts of modeling memory, How to model memory and initialize memory, concepts of modeling register bank	3
9	**Verilog modeling of processor** RISC Processor, Basic pipe line concepts, Understanding pipe line implementation	4
10	**Programmable Logic Devices** Introduction to Programmable Logic Devices	2
Total Hours		**42**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Experiment-1** Write the Verilog code to implement given logic functions.	2
2	**Experiment-2** Write the Verilog code from the given logic circuit.	2
3	**Experiment-3** Write the Verilog code for designing 16x1 Multiplexer with 2x1 multiplexer.	2
4	**Experiment-4** Write the Verilog code for designing 4x16 decoder with 2x4 decoder.	2
5	**Experiment-5** Write the Verilog code for designing 8 to 3 priority encoder.	2
6	**Experiment-6** Write the Verilog code for designing half adder and full adder circuit.	2
7	**Experiment-7** Write the Verilog code for designing 4 bit ripple carry adder circuit.	2
8	**Experiment-8** Write the Verilog code for designing binary to gray code converter circuit.	2

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
9	**Experiment-9** Write the Verilog code for designing BCD to seven segment converter circuit.	2
10	**Experiment-10** Write the Verilog code for D latch, SR latch and JK latch.	2
11	**Experiment-11** Write the Verilog code for D flip flop, SR flip flop, JK flip flop.	2
12	**Experiment-12** Write the Verilog code for designing T-flip flop using D-flip-flop	2
13	**Experiment-13** Design 4-bit shift register with asynchronous clear and enable using verilog.	2
14	**Experiment-14** Design 4-bit shift register with synchronous reset using Verilog.	2
15	**Experiment-15** Design 4 bit up/down counter with parallel load using Verilog.	2
16	**Experiment-16** Design Moore machine(FSM) with example using Verilog.	2
17	**Experiment-17** Design Mealy machine(FSM) with example using Verilog.	2
18	**Experiment-18** Design 4 bit ALU which performs the following tasks. a. Add b.Subtract c.Multiplication d. Division e. OR f. AND g. Invert h.XOR	2
19	**Experiment-19** Implement bus structure to transfer data among registers using two different approaches. a. Using tristate driver b. Using multiplexer	2
20	**Experiment-20** Design control circuit for a simple processor to perform basic operations mentioned below. a. load data into register b. copy data from register to register c. add d. subtract	2
Total Hours		**40**

Textbook :

1 Verilog HDL Synthesis-A practical primer, J.Bhaskar, star galaxy, 1999

2 Verilog Digital System Design, Z.Navabi, McGraw-Hill Education, 2005

3 Digital Design: Principles and practices, Jon F. Wakerly, Prentice Hall, 1990

References:

1 Verilog HDL A guide to digital design and synthesis, Verilog HDL A guide to digital design and synthesis, Samir Palnitkar, Pearson Education, 2003

2 Fundamentals of Digital Logic with Verilog Design, Fundamentals of Digital Logic with Verilog Design, Stephen Brown and Zvonko Vranesic, McGraw-Hill, 2002

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
5.00	20.00	35.00	30.00	0.00	10.00

Instructional Method:

1. Course delivery is combined use of white board/black board and power point presentation using projector as per requirement.
2. Students will be assessed with continuous evaluation approach with assignments and quizzes at regular interval.
3. For effective communication with students, canvas LMS (Learning Management System) is used.
4. Students are given opportunity to interact with mentors from industries through mock interview.

Supplementary Resources:

1. Hardware Modeling Using Verilog: Prof.Indranil Sengupta, IIT Kharagpur https://nptel.ac.in/courses/106/105/106105165/
2. Digital System Design with PLDs and FPGAs: Prof. Kuruvilla Varghese, IISc Bangalore https://nptel.ac.in/courses/117/108/117108040/
3. https://www.edaplayground.com/

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